Table 17 Within-distribution performance comparison: Debate-derived majority vote supervision vs. ground truth supervision on training domains. Models are trained on MATH, GSM8K, and MathQA datasets. Bold indicates the better method for each model-dataset pair, demonstrating that debate-derived supervision achieves comparable performance to ground truth labels.

Model	Method	MATH		GSM8K		MathQA	
		Debate-MV	GT	Debate-MV	GT	Debate-MV	GT
Qwen2B	Base	10.4	10.4	27.0	27.0	7.4	7.4
	SFT	**10.8**	10.4	25.6	**26.4**	8.2	**8.8**
	GRPO	19.4	**21.0**	45.2	**48.6**	18.6	**19.6**
	KTO	22.6	**23.2**	**54.8**	54.6	**28.8**	28.6
	DPO	**24.8**	24.2	51.4	**52.0**	**24.2**	24.0
Llama3B	Base	32.0	32.0	69.6	69.6	24.6	24.6
	SFT	**33.2**	32.4	64.2	64.2	**26.4**	25.2
	GRPO	45.8	**46.4**	**75.8**	74.8	**36.2**	31.8
	KTO	**48.0**	47.8	76.0	**76.8**	**41.4**	40.6
	DPO	53.2	**53.6**	**80.4**	77.8	**46.2**	45.4

Table 18 Cross-domain generalization: Debate-derived majority vote supervision vs. ground truth supervision on unseen reasoning tasks. Models trained on mathematical datasets (MATH, GSM8K, MathQA) generalize effectively to diverse reasoning domains. Bold indicates the better method for each model-dataset pair, showing that both supervision approaches transfer well to out-of-distribution tasks.

Model	Method	SVAMP		GPQA		CSQA	
		Debate-MV	GT	Debate-MV	GT	Debate-MV	GT
Qwen2B	Base	48.30	48.30	0.45	0.45	3.80	3.80
	SFT	**53.30**	53.00	**17.90**	0.89	16.80	**18.80**
	GRPO	**60.30**	58.33	**8.70**	7.81	**10.80**	9.60
	KTO	76.00	**76.33**	16.70	**16.96**	**63.40**	60.80
	DPO	**65.00**	64.67	19.64	**20.98**	**62.2**	60.80
Llama3B	Base	71.30	71.30	0.67	0.67	53.00	53.00
	SFT	68.33	**72.0**	**2.23**	2.23	57.40	**57.80**
	GRPO	75.00	**79.33**	**6.92**	5.13	**63.20**	59.40
	KTO	76.70	**78.67**	8.93	**9.82**	**62.20**	61.80
	DPO	78.40	**80.67**	**11.40**	11.60	**64.00**	62.40

ground truth supervision, achieving comparable performance without human annotation. This approach offers significant advantages for scaling reasoning improvements to new domains or large datasets where expert labels are unavailable or prohibitively expensive.

K Post-Training Method Impact on Log-Probability Distribution

Figure 6 shows how different post-training methods affected log-probability distributions for Qwen-2B on MATH. Preference-based methods (DPO, KTO) increased the density of majority distributions, with KTO showing particularly pronounced effects: higher peaks and tighter tails indicating more concentrated probability mass around consensus responses. SFT left the majority distribution largely unchanged while substantially reducing minority distribution density. GRPO created the most dramatic separation between distributions, shifting both leftward (lower log-probabilities overall) but with the minority distribution shifting much more substantially than the majority. While these different patterns (probability concentration for preference methods versus selective penalization for GRPO) all corresponded with performance improvements, further research is needed to better understand the relationship between these specific distributional changes and the impact on consistency.

L Preference Learning as an Implicit Format Reward

This section investigates how preference learning through consensus signals acts as an implicit format reward, teaching models to produce more efficient and accurate reasoning without explicit formatting supervision. We examine three key

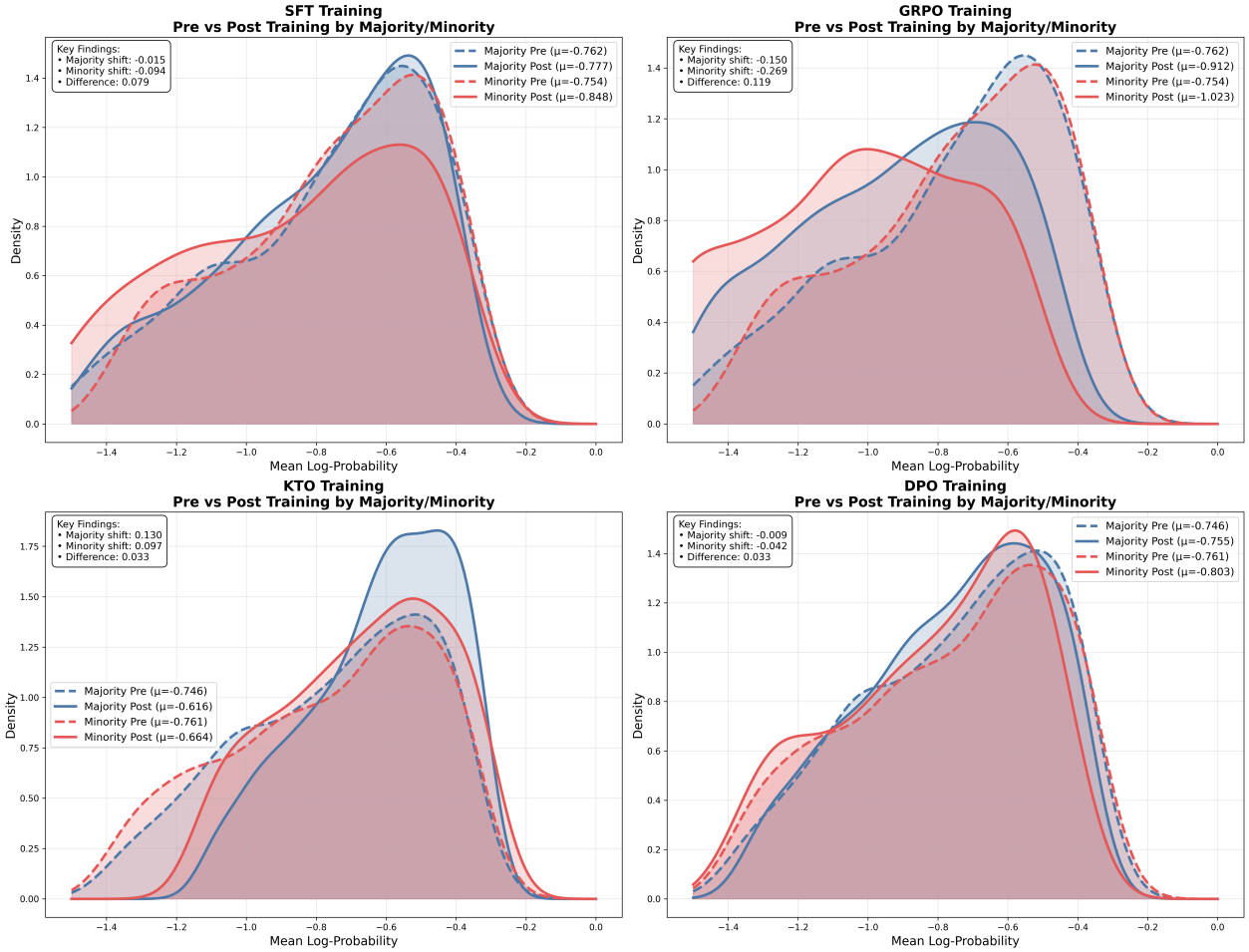

Figure 6 Log probability distributions for majority and minority answers before and after post-training (Qwen2B, Math).

aspects: the baseline capability requirements for effective consensus formation, how token constraints interact with reasoning improvements, and the decomposition of performance gains into format improvements versus problem-solving accuracy improvements. Crucially, we demonstrate that while preference learning does function effectively as an implicit format reward, most of the performance gains observed are attributed to fundamental improvements in problem-solving accuracy rather than mere formatting compliance.

L.1 Baseline Capability Requirements for Consensus Formation

Our experiments reveal that MACA requires sufficient baseline model capability to generate meaningful consensus signals. When models lack foundational problem-solving ability, they fail to produce the correct responses necessary for consensus-based reinforcement. Table 19 illustrates this limitation: under a strict 256-token limit, Qwen2B produces no correct responses on AMC, resulting in no useful consensus to reinforce. Increasing the token limit to 512 partially mitigates this by allowing more reasoning space, though inefficient reasoning patterns can still cause truncation.

Table 19 Impact of token limits and baseline capability on AMC performance. Shows accuracy percentages for base models and post-trained models ("All") under different token constraints. When models lack sufficient baseline capability (Qwen2B at 256 tokens), consensus formation fails as no correct responses are generated for reinforcement.

	max_new_tokens = 256		max_new_tokens = 512	
	Base	All	Base	All
Qwen2B	0.0	0.0	5.0	**12.5**
Llama3B	7.5	**10.0**	10.0	**20.0**

This baseline capability requirement has important implications for applying MACA: models must possess some initial problem-solving ability on the target domain to benefit from consensus-based training. However, once this threshold is met, we observe that improvements on easier datasets can generalize to more challenging tasks, suggesting that enhanced self-consistency helps overcome some limitations of consensus formation on difficult problems.

L.2 Token Efficiency and Reasoning Quality Improvements

Post-training simultaneously addresses token efficiency and reasoning quality. Models trained with debate under constrained token limits generate more parseable answers within the budget while achieving higher accuracy on those answers. To verify that improvements extend beyond mere format optimization, we evaluated models post-trained with 256-token debates using 512-token test budgets.

Table 20 demonstrates that performance gains persist under increased token limits, indicating genuine reasoning enhancement rather than just format improvements. We also analyze the sequence lengths of the trajectories generated for the analysis in App. I, showing that uncapped post-trained models generate responses that are 22-36% shorter than their base models. This evidence suggests that self-guided preference learning functions as an implicit format reward, teaching models to produce more concise and effective reasoning patterns that generalize across computational budgets and task difficulties.

L.3 Decomposing the Sources of Improvement

To understand whether our gains stem from improved reasoning or merely better formatting, we decompose performance improvements into their constituent components. Our analysis separates gains from better completion (avoiding truncation) versus fundamental reasoning improvements.

The results reveal that 69-100% of improvements stem from better reasoning rather than just avoiding truncation. While base models exhibit high truncation rates (e.g., 74.8% for Qwen2B on CSQA, 82.8% on MATH), post-training reduces these dramatically. However, even after accounting for completion improvements, substantial reasoning gains remain, confirming that our method teaches fundamentally better problem-solving, not merely more efficient token usage.